

4. Heat Exchanger Tube Bundle Retirement Thickness Evaluation

To: Mechanical Integrity Department, Operations **From:** A. Reyes, Mechanical Integrity Engineer **Date:** 2026-06-24 **Subject:** Exchanger E-305 — Tube Bundle Retirement Thickness Result (per SOP-HX-160 Clause 4.6)

Background

Per SOP-HX-160 Section 4, Clause 4.6, tube bundle retirement wall thickness must be recalculated following any eddy-current inspection showing degradation exceeding the Clause 4.8 engineering-review threshold. The turnaround eddy-current survey on E-305's shell-side bundle identified wall loss at multiple tube support locations meeting this trigger.

Assessment Result

The calculated minimum retirement thickness for E-305 tubing, based on design pressure, tube OD, and material allowable stress, is **0.71 mm**. This value supersedes the generic bundle retirement figure referenced in the prior turnaround scope document.

Any tube with eddy-current-indicated wall thickness below 0.71 mm shall be plugged prior to bundle re-installation, and the bundle shall not be returned to service with more than 10% of tubes plugged without a documented thermal derate review per SOP-HX-160 Clause 4.9.

Distribution

Filed to the Heat Exchanger Integrity File per SOP-HX-160 Clause 6.3. Retain for the operating life of the bundle plus 10 years.